

BIOLOGY BOARD QUESTION PAPER : JULY 2025**Time: 3 Hrs.****Max. Marks: 70****General Instructions:**

1. The question paper is divided into four sections.
2. Section A: Q. No. 1 contains ten multiple choice type of questions carrying one mark each. Evaluation will be done for the first attempt only.
3. Section A: Q. No. 2 contains eight very short answer type questions carrying one mark each. (Attempt any Six)
4. Section B: Q. 3 to 14 are short answer type questions carrying two marks each. (Attempt any Eight)
5. Section C: Q. 15 to 26 are short answer type questions carrying three marks each. (Attempt any Eight)
6. Section D: Q. 27 to 31 are long answer type questions carrying four marks each. (Attempt any Three)
7. Begin the answer of each section on a new page.

SECTION – A

Q.1 Select and write the correct answer for the following multiple-choice type of questions: [10]

- i. The number of base pairs present around a histone octamer of nucleosome is
(A) 200 (B) 146 (C) 54 (D) 20
- ii. Which of the following is NOT a sensory nerve?
(A) I (B) II (C) III (D) VIII
- iii. Which of the following biofertilizers is used by farmers to increase the yield of rice crop?
(A) Azotobacter (B) Azospirillum (C) Bacillus (D) Azolla
- iv. Connecting link between amphibians and reptiles is
(A) Archaeopteryx (B) Australopithecus (C) Seymouria (D) Ichthyostega
- v. The type of gene interaction deviating from Mendel's findings 3 : 1 to 1 : 2 : 1 is
(A) Incomplete dominance (B) Codominance (C) Multiple alleles (D) Pleiotropy



vi. Identify initiation codon of mRNA:

(A) UAA (B) AUG (C) UAG (D) UGA

vii. Identify the type of association between orchid growing as an epiphyte on branch of mango tree.

(A) Parasitism (B) Amensalism (C) Commensalism (D) Mutualism

viii. Which of the following substances protects pollen grains from chemicals?

(A) Pectin (B) Hemicellulose (C) Cellulose (D) Sporopollenin

ix. The part of antibody which binds with the specific antigen to form antigen–antibody complex is

(A) Light chains (B) Isotopes (C) Epitopes (D) Antigen

x. Which genes are responsible for improvement of oil content and oil quality in plants?

(A) Flax (B) Arabidopsis spp. (C) Linseed (D) Oncogene

Q.2 Answer the following: [8]

i. Give term for: “The total genetic information encoded in sum total of genes in a Mendelian organism.”

ii. Identify the following complementary base pairing:

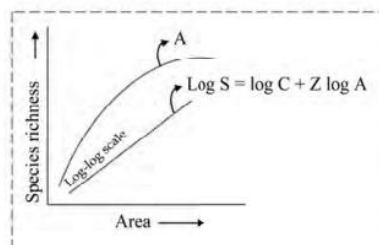
3' CTT AAG

5' GAA TTC

Give the term.

iii. What is the term when nucleotides on DNA strands show this type of base pairing?

iv. Identify 'A' from the given species–area relationship graph.



v. Name the disorder in which deficiency of ADH leads to reduced water reabsorption and increased urine output.

vi. Name the permanent birth control method in which fallopian tube is tied and cut.

vii. Name the long-term storage places of carbon in form of rocks and fossil fuels which are away from carbon cycle.

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SECTION – B**(Attempt any EIGHT)[16]**

Q.3 Give any two reasons why breathing through nose is beneficial than through mouth.

Q.4 Mention any two molecular evidences of evolution.

Q.5 Enlist any four uses of gene therapy.

Q.6 Define the terms:

i. Explant

ii. Callus

Q.7 Identify the disorders showing the following symptoms:

i. Constriction of bronchioles, mucus build-up, periodic wheezing, difficulty in breathing.

ii. Inflammation of sinuses and mucous discharge.

Q.8 Define AGR and RGR regarding plant growth.

Q.9 Draw neat labelled diagram of root hair showing vacuole, mitochondria, nucleus.

Q.10 Enlist the types of chromosomes on the basis of position of centromere.

Q.11 Mention any four drawbacks of Darwinism.

Q.12 Write appropriate equations to represent Ammonification.

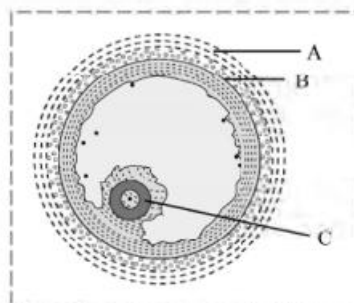
Q.13 Give any two functions of Medulla oblongata.

Q.14 Differentiate between Inspiration and Expiration with respect to diaphragm position and working of intercostal muscles.

SECTION – C**(Attempt any EIGHT)[24]**

Q.15 What is adolescence? Mention any four characteristics of 15 to 17 years age group.

Q.16 Identify A, B, C from the given diagram of Graafian follicle and mention their significance.



Q.17 i. Which part of CNS produces CSF?

ii. Give any four functions of CSF.

Q.18 Write the name and position of any three valves in human heart.

Q.19 Describe any three chemical defence mechanisms in plants against herbivores in population interaction.

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Q.20 Mention any six symptoms of disorder caused by trisomy of 21st chromosome.

Q.21 Describe the development of male gametophyte from microspore. Draw a neat labelled diagram of male gametophyte.

Q.22 Define decomposition. Explain any four steps of decomposition in correct sequence.

Q.23 Give deficiency symptoms of following elements in plants:

i. Potassium (K^+)

ii. Magnesium (Mg^{2+})

iii. Boron (BO_3^{3-})

Q.24 Give scientific names of sugarcane varieties native of North and South India. Name the centre in India where high-sugar varieties were developed.

Q.25 Complete and rewrite the following chart:

Sr. No.	Disorder/Diseases	Recombinant Protein(s)
i.	Parturition	
ii.	Emphysema	
iii.		Hepatitis B

Q.26 Mention any two major types of air pollutants. Add a note on their ill-effects on human health.

SECTION – D

(Attempt any THREE)[12]

Q.27 Define endosperm. Describe its types.

Q.28 With the help of a neat labelled diagram, describe the structure of human sperm.

Q.29 Describe the transcription and processing of hnRNA to mRNA in eukaryotes.

Q.30 What is transpiration? Draw labeled diagram of stomatal apparatus in dicot leaf. Mention any three advantages of transpiration.

Q.31 Name any four hormones secreted by adenohypophysis and give their significance.

